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Summary

ECE MASC engineer (Queen's University, Dean's Scholar) working across hardware design, embedded systems, digital signal processing, applied machine learning and computer vision, robotics, RTL and FPGA design, and medical instrumentation. Approximately three years of hands on paid experience across applied research at NavINST (Queen's and RMC), part time PCB and firmware engineering at IoBIO Inc., and co founding a two person medical imaging startup. Personal RTL portfolio in Verilog and VHDL targets real time video and DSP pipelines, including a single cycle RV32I RISC V CPU core and a per subpixel routing IP block for an autostereoscopic display, each with self checking testbenches and end to end simulation. Additional experience in Zephyr RTOS, PyTorch and TensorFlow, GTSAM factor graph fusion, CUDA accelerated real time processing under Linux, polarimetric imaging, and custom RF antenna design. Documentation and interdisciplinary habits developed across multi team hardware, firmware, algorithms, and product deliveries.

Skills

- Circuit and PCB Design: schematic capture and multilayer mixed signal PCB layout in Altium Designer and KiCad; analog signal chain and low noise power design; microcontroller based products (ESP32-S3); component selection across ICs and passives; iteration across board revisions from bring up and supply findings; custom RF antenna design for satellite platforms.
- Bring Up, Validation, and Debug: hardware bring up with oscilloscope, logic analyzer, DMM, function generator, and JTAG; power supply and signal integrity validation; board and system level debugging; documented rationale and test results at every revision; coordination with fabrication and assembly vendors on Gerbers, stack up, and DFM feedback.
- Embedded Systems and Firmware: register level device drivers in C and C++ (I2C, SPI, UART, GPIO); FreeRTOS interrupts and hand off patterns from ISR to task; working Zephyr RTOS experience (k_msgq, k_thread, K_THREAD_STACK_DEFINE, K_MSGQ_DEFINE, logging subsystem, west workspace, device tree overlays, Kconfig); C11 atomics with acquire and release ordering; lock free single producer and single consumer ring buffers; byte at a time framing state machines with CRC validation; onboard DSP and networked telemetry.
- Digital Signal Processing: onboard DSP, filter design, sensor fusion algorithms; beamforming and image reconstruction from raw echoes; CUDA accelerated real time processing.
- Digital Design and FPGA: Verilog, VHDL, RTL design of video and display pipelines and high throughput datapaths; self checking testbenches; module and system level verification and validation strategies; fixed point (Q format) arithmetic; pipelined datapath design; AXI Stream interfaces; simulation with Icarus Verilog, GHDL, and GTKWave; targeting AMD Xilinx Vivado toolchain and Basys 3 and Zynq 7000 platforms; ongoing TinyTapeout self study covering OpenLane, floorplanning, place and route, and timing closure.
- Machine Learning, Computer Vision, and Sensor Fusion: PyTorch and TensorFlow for ML pipelines; OpenCV for classical CV; GTSAM factor graph fusion (IMU preintegration, GPS factors, custom polarization yaw factor with confidence scaled noise); polarimetric imaging including Stokes parameters, DoLP, and AoLP in IPP and SPP coordinate systems; ROS 2 multi sensor integration.
- Robotics and Simulation: end effector and gripper design for the Franka Emika Panda arm; motion control on Raspberry Pi and microcontroller platforms; NVIDIA Isaac Sim digital twin development for validating grasping behaviours and motion sequences prior to hardware deployment; SolidWorks FEA validated mechanics.
- Power and Analog: onboard low noise LDO rails feeding precision ADC front ends; analog signal conditioning; graduate level modelling and control of switching power converters.
- Languages, Tools, and Version Control: C, C++, Python, JavaScript; Linux development environment; Git and GitHub with code review as daily practice; Jira and Asana for planning and tracking; AddressSanitizer, UBSan, and -Wall -Wextra -Werror -Wpedantic clean builds; plain assert unit testing; ztest for Zephyr; daily use of modern AI assistants (Claude, Gemini) for research, analysis, and engineering tool development.
- Test, Debug, and Documentation: unit, integration, and system testing with sanitizers and warning clean builds; oscilloscope, logic analyzer, DMM, signal generator; design rationale and test result documentation for interdisciplinary review; project lead on subprojects at NavINST and Queen's Microelectronics and Photonics Group, mentoring undergraduate students and interns and reporting progress to the principal investigator.

Work Experience

Research Assistant

September 2025 to Present

Calian Group Ltd. (NavINST Research Group, Queen's University and Royal Military College)

Kingston, Ontario

- Deliver real time estimation on live multi sensor navigation streams by architecting a Python and C data processing pipeline under Linux, with throughput critical kernels offloaded to CUDA on GPU; instrument stack includes camera, NovAtel GNSS/INS, u-blox GNSS, and Xsens IMU streaming through a ROS 2 stack.
- Built and validated machine learning and classical computer vision algorithms in PyTorch, TensorFlow, and OpenCV for a polarization based heading estimator, including raw frame demosaicing, Stokes S0, S1, and S2 computation, DoLP and AoLP in IPP and SPP coordinate systems, four camera affine alignment, and interactive polygon and brush masking tools.
- Developed a GTSAM based polarization aided SLAM pipeline in Python that fuses IMU preintegration, GPS factors, and a custom polarization yaw factor with confidence scaled noise; benchmarked yaw accuracy against a survey grade NovAtel reference (approximately 0.35 degrees RMSE for the deep learning fusion output, and approximately 0.794 degrees for the standalone polarization compass).
- Fused the polarization compass yaw solution with Xsens roll and pitch, which reduced heading error from approximately plus or minus two degrees to between 0.1 and 0.5 degrees across weather conditions, benchmarked against the NovAtel reference; surfaced a distribution shift failure mode through per minute error plots and addressed it with a DoLP scaled noise model.
- Serve as project lead and daily supervisor on assigned subprojects, mentoring undergraduate students and interns and reporting progress to the principal investigator; previously performed the same role in Queen's Microelectronics and Photonics Group.
- Keep hardware, embedded, and algorithms teams aligned on interdisciplinary deliverables by producing documented, reviewable design and test artifacts at each integration step.

Founder

May 2025 to Present

WaveWise Health

Kingston, Ontario

- Founded and lead a two person medical imaging startup building an AI guided handheld cardiac ultrasound device for coronary flow assessment; own hardware, embedded, and machine learning implementation end to end.
- Took the ultrasound front end from specification through functional bring up by designing the transducer driver and interface PCB in Altium and the low noise analog signal chain feeding a high speed digitizer, working to functional and safety expectations for a medical device.
- Reconstructed diagnostic images from raw echoes by implementing onboard DSP in C and C++ for signal conditioning and beamforming, validated against reference phantoms.
- Developing machine learning models to predict heart related conditions from ultrasound derived signals, iterating on data collection strategy, model architecture, and validation against reference standards.
- Sourced components and managed supplier timelines against build schedules for prototype iterations.
- Secured seed stage R&D funding by presenting technical progress, architecture, and roadmap to early investors.

Electrical Engineer (Part time)

May 2023 to February 2026

IoBIO Inc.

Kingston, Ontario

- Delivered a microcontroller based industrial IoT sensor product across multiple board revisions by designing custom multilayer mixed signal PCBs in KiCad and Altium around the ESP32-S3, integrating a 16 bit ADC on a dedicated low noise LDO rail for precision analog acquisition, and taking the design from schematic capture and layout through BOM and assembly.
- Enabled reliable end to end data capture and transport by developing embedded firmware in C and C++, including register level I2C sensor drivers, onboard DSP for signal conditioning, a FreeRTOS task and queue architecture, and networked Wi-Fi telemetry into a cloud backend that enables real time customer monitoring of field deployed sensors.
- Reduced respin risk across revisions by executing structured board bring up and power supply and signal integrity validation with oscilloscope, logic analyzer, DMM, and JTAG, then folding findings into schematic and layout changes.
- Coordinated with fabrication and assembly vendors on Gerbers, stack up, and DFM feedback across revisions, and worked with product stakeholders on form factor and interface constraints.
- Enabled hand off across hardware, firmware, and quality collaborators by documenting design rationale, schematic and layout deltas, and test results at every revision.

- Used AI assistants (Claude, Gemini) daily as part of design, debug, and test case generation to increase firmware quality and throughput.

Electronics Engineering Intern

KDU Worldwide Technical Services Gh. Ltd.

May 2019 to August 2019

Tema, Ghana

- First hands on FPGA and embedded exposure implementing basic logic in support of field equipment including power supplies and signal repeaters.
- Returned field equipment to service by diagnosing faults to the component level with oscilloscope and DMM and executing board level repairs.
- Prevented repeat failures on redeployed equipment by verifying fixes against acceptance criteria and documenting repair procedures for the service team.

Projects

Human Robot Interaction End Effector, Franka Emika Panda (ELEC 490 Capstone) *September 2022 to April 2023*

- Delivered a full electromechanical end effector for the Franka Emika Panda 7 DOF arm designed for human robot interaction: owned the mechanical design in SolidWorks with FEA validated mechanics, the electrical design (custom control PCB with actuator drivers and sensor interfaces), and the real time firmware and motion control software running on a Raspberry Pi 5 controller.
- Built an interactive Franka Panda plus end effector digital twin model in NVIDIA Isaac Sim, using the simulator to validate grasping behaviours and motion sequences against the physical rig before hardware deployment.
- Exceeded specification by holding 7.2 kg against a 2 kg payload target at approximately \$300 in parts, with the gripper safe for human interaction in a shared workspace.

RV32I RISC V CPU Core (Verilog, Personal)

2025 to Present

- Designed and verified a single cycle RV32I RISC V processor in Verilog, implementing the full base integer ISA (approximately 37 instructions across R, I, S, B, U, and J type encodings) with ALU, branch unit, and load and store path.
- Architected instruction and data memory interfaces with a 32 register file, hardwired zero x0, and combinational next PC logic supporting conditional branches, JAL, and JALR.
- Verified end to end by developing a self checking Verilog testbench with per cycle instruction trace and automated pass and fail reporting, executing hand assembled RV32I machine code and checking results in the data memory.
- Established a portable, tool agnostic verification flow by simulating in Icarus Verilog with waveform capture in GTKWave.

Multi View Pixel Router for Autostereoscopic Display (VHDL, Personal)

2025 to Present

- Designed a real time, per subpixel pixel routing IP block in VHDL for an 8 view slanted lenticular 3D display, implementing van Berkel's canonical view selection formula in Q16.16 fixed point arithmetic.
- Sustained one output pixel per clock by architecting a 2 stage pipelined RTL datapath with three independent view index calculators, a parallel 8 way memory read, and a 3 way subpixel mux for independent R, G, and B routing.
- Enabled reuse across panel configurations without RTL changes by parameterizing lens pitch, tilt, view count, and frame dimensions as generics.
- Verified correct routing end to end by building a VHDL 2008 self checking testbench modelling 8 external view memories with 1 cycle read latency, visualizing the interleave pattern, and comparing pixel counts against expected frame size.
- Simulated in GHDL; documented trade offs between BRAM and throughput and between combinational and pipelined divide, and scoped a synthesis path including reciprocal multiply optimization and an AXI Stream wrapper.

zephyr-uart-framing-node: Zephyr RTOS Port of ringbuf-parser (Personal)

2026

github.com/Deryanshagarwal/zephyr-uart-framing-node

- Ported the C11 ringbuf-parser framing library to Zephyr RTOS, replacing lock free atomics with k_msgq for the hand off from ISR to thread; standard Zephyr application layout including a west manifest, prj.conf, Kconfig, device tree overlays, and the Zephyr logging subsystem.
- Six case ztest suite covering well formed frames, partial frame delivery, garbage resync, CRC error recovery, back to back frames, and the CRC-16/CCITT-FALSE golden vector.
- GitHub Actions CI targeting native_sim through the official Zephyr container.

ringbuf-parser: Embedded Byte Stream Framing Library (C11, Personal)

2025 to Present

github.com/Devianshagarwal/Ring-buffer-SPI-I2C-protocol-parser-in-C

- Designed a single producer and single consumer lock free ring buffer using C11 atomics with acquire and release ordering, modeling the hand off pattern from ISR to task used on UART and SPI receive paths.
- Implemented a byte at a time framing state machine with CRC-16/CCITT-FALSE validation, automatic resync on corruption, and telemetry counters for diagnostics.
- Built plain assert unit tests covering wrap around, partial frame delivery, garbage resync, and CRC error paths; compiles clean under -Wall -Wextra -Werror -Wpedantic and passes AddressSanitizer and UBSan.

Polarization Aided SLAM Fusion Pipeline (Python, GTSAM, Personal Research)

2026 to Present

- Built a factor graph fusion pipeline in Python integrating IMU preintegration, GPS factors, and a custom polarization yaw factor with DoLP scaled noise.
- Run against a dataset of 50 stationary and dynamic polarization image pairs from an approximately six minute vehicle collection run, each pair tagged with NovAtel roll, pitch, and yaw, odometry velocity, and IMU measurements (NovAtel and Xsens), with pair time offsets between 0.5 and 97 milliseconds.
- Closes the loop from live vehicle collection data back to refined models, with per segment error analysis surfacing distribution shift failure modes and driving iteration on the noise model.

Polarimetric Imaging Software Pipeline (Python, OpenCV, matplotlib, Personal Research)

2025 to Present

- Iteratively developed a computer vision pipeline for the PHX050S-P polarization camera covering raw frame demosaicing via polanalyser, Stokes S0, S1, and S2 computation, and DoLP and AoLP computation in IPP and SPP coordinate systems with a beta angle matrix for SPP rotation.
- Built a four camera alignment GUI with per camera affine transforms and AoLP SPP output, alongside interactive annotation tools including a magic brush masking widget with jet colormap DoLP histograms and a polygon widget for NaN based object isolation.
- Built yaw analysis and RMSE plotting scripts with per minute annotations against auto calibrated reference data, and standardized display conventions including DoLP colormap fixed at vmin 0 and vmax 1 and AoLP fixed at vmin minus 90 and vmax 90.

Custom RF Antenna Design for CubeSat Program (Queen's University)

Undergraduate Research

- Designed custom RF antennas for a Queen's CubeSat program under Dr. Siddiqui, working through antenna geometry, impedance matching, and simulation to fabrication trade offs for a size and weight constrained satellite platform.

Real Time Attitude Estimator on ESP32 and FreeRTOS (Personal)

In Progress

- Sustained a 200 Hz sensor fusion loop with measured timing performance by implementing a register level I2C IMU driver and a dual core FreeRTOS task and queue architecture in a Linux based development environment.

In Browser Digital Logic Simulator (JavaScript, Personal)

September 2024 to Present

- Built an interactive digital logic simulator in JavaScript with waveform rendering and preset designs to visualize combinational and sequential circuit behaviour in the browser.

Education

Master of Applied Science, Electrical and Computer Engineering

September 2023 to 2025

Queen's University

Kingston, Ontario

- Graduate coursework: Advanced Power Electronics covering modelling and control of switching power converters, Antenna Engineering, Introduction to Nanophotonics, and Uncenventional Computing.

Bachelor of Applied Science, Electrical and Computer Engineering (Dean's Scholar)

September 2019 to April 2023

Queen's University

Kingston, Ontario

Continuing education and certifications: *Introduction to VHDL for FPGA and ASIC Design (Scott Dickson); ongoing self study of the open source ASIC and FPGA flow through TinyTapeout (OpenLane: floorplanning, place and route, timing closure) and online courses on synthesis, static timing analysis, logical equivalence checking, and clock domain crossing.*